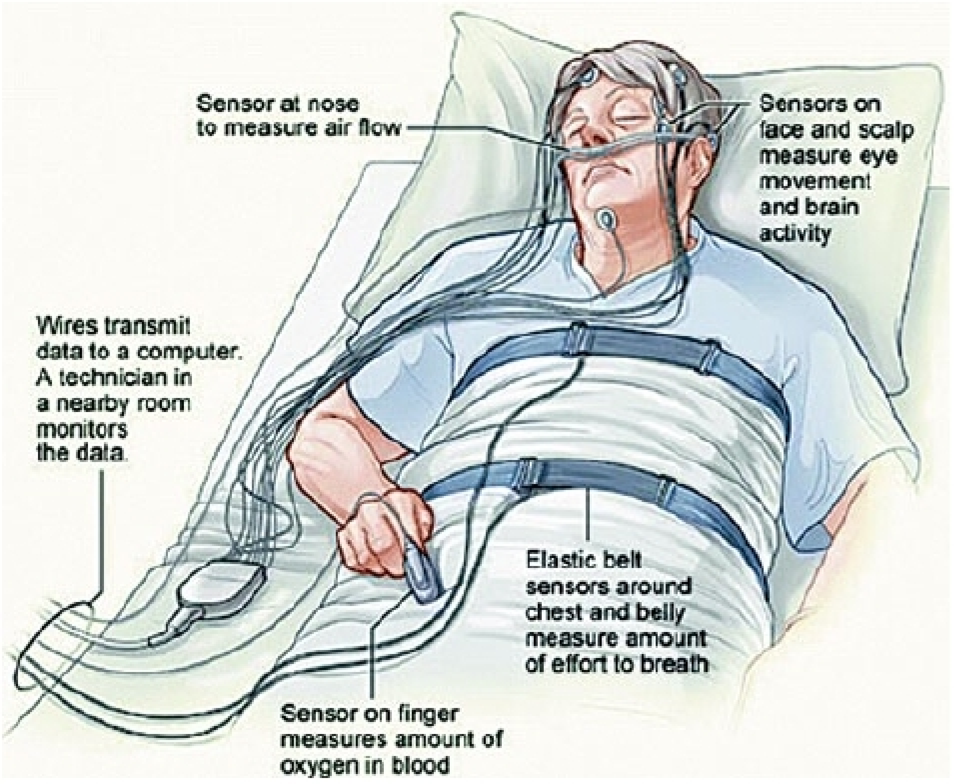
device. There is always a trade-off between device performance and battery life for the intended period of use. Advancements are being made for utilising biologically generated energy, or energy generated by body movements.

Other challenges. Besides physical and technical, some other challenges are:

- The method used for providing alerts and feedback—visual, auditory or tactile—be relevant, comprehensible and reliable. As the stored medical information is personal and private, it should be accessible only to authorised users. Encryption techniques and levels are appropriate, reliable and balanced with the power capacity of the device.
- The device be user-friendly, especially when in use by inexperienced or uneducated users, to ensure safety of the patient during usage without causing any harm to the patient, or creating nuisance and inconvenience (such as skin rashes).
- Reliability of outputs and results is an important factor. Non-controlled environment such as varying humidity and ambient temperature, and physical factors such as sweating, vibration, or body movements should not affect reliability.
- Devices should comply with applicable standards, codes and regulations to ensure their acceptability, safety and reliability. International standards are available such as ISO 11073 and IEEE Standard 1073.
- These must meet legal, regulatory and ethical requirements.
- Costs including professional fee, cost of emergency handling, training to be imparted to the patient to setup and use, and the maintenance, should be within the easy reach of patient and commercially viable.



A polysomnograph device in use (Credit: www.sleep-apnea-guide.com)

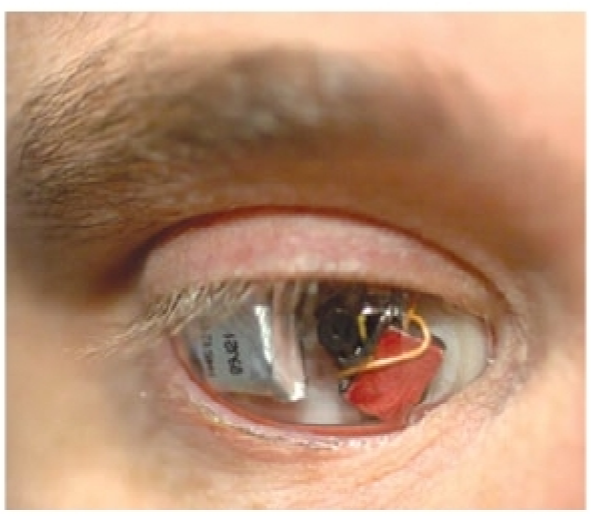
Applications

A range of wearable medical devices are readily available for use by patients and healthcare providers. These can be divided broadly into two categories: medical monitoring devices and medical rehabilitation devices.

Wearable medical monitoring devices. These devices are designed for non-invasive sensing, monitoring vital medical parameters, communicating with patients, and providing feedback and instant alerts, or medical decisions to patients and medical supervisors. Advanced devices monitor athletes during exercise and performance, and patients within the healthcare environment. A few examples are:

Wearable armband. A wireless body monitor worn on the arm for collecting physiological and lifestyle data such as movements made, heat flow, skin temperature, heart rate, sleep logs, etc.

Vivometrics LifeShirt. A non-invasive ambulatory monitoring garment with embedded sensors, which is used to monitor up to forty such physiological parameters as heart rate, breathing and lung capacity.



A man wearing a specialised camera for an eye (Credit: http://s.telegraph.co.uk)

Polysomnograph (PSG). Also called sleep tracker, PSG is a wearable multi-channel data recorder body patch worn on the chest of a patient to record ECG, EMG, electrooculography, EEG, chest effort, pulse oxymetry, body position, airflow, etc.

Wearable glucose monitor. A wrist-worn, non-invasive blood glucose monitoring device used by diabetic patients to provide readings every fifteen minutes by extracting glucose through skin via reverse iontophoresis.

Stress monitoring device. A wearable device used by drivers for determining stress levels while driving by monitoring/measuring parameters such as skin conductance, respiration, muscle and heart activity, and capturing facial expressions.

Wearable device for athletes. A device used for real-time monitoring of athletes' movements, changes in position of their body, respiratory effort, heart condition, temperature, blood pressure and detail of any injury sustained. Real-time feedback is available to athletes and coaches via a virtual reality (VR) interface, which enables them to improve performance.

Wearable medical rehabilitation devices. These devices provide long-term support, medical aid and rehabilitation assistance to patients with temporary or permanent disabilities. Some representative examples are:

Rehabilitation device for nerve stimulation. It is used for therapeutic purposes to block pain by providing functional stimulation through electrical impulses to the patient.

Wearable motion analysis device. The device is used as a diagnostic tool as well as a biofeedback device during rehabilitation therapy to monitor patients with balance disorders. It is also useful as a fall-prevention aid for fall-prone elderly persons. Triaxial digital accelerometers are attached on an eyeglass frame for measuring